

SAGAR S

Bangalore, India

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PROFESSIONAL SUMMARY

Machine Learning Engineer with **1.8 years of experience** building and deploying AI/ML systems focused on **LLMs, Retrieval-Augmented Generation (RAG), NLP, and MLOps**. Experienced in developing scalable AI applications, fine-tuning models, building backend APIs, and deploying production-ready ML systems on Google Cloud. Strong expertise in Python, vector databases, model optimization, and cloud-native deployment.

TECHNICAL SKILLS

Programming: Python, SQL, Java

Machine Learning & AI: Machine Learning, Deep Learning, NLP, LLMs, Generative AI, RAG, Prompt Engineering, Fine-tuning (LoRA/PEFT), Model Evaluation

Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, LangChain, LlamaIndex, FastAPI

Cloud & DevOps: Google Cloud, Docker, Kubernetes, CI/CD, GitHub Actions, MLflow

Databases: PostgreSQL, MongoDB, Redis, FAISS, Pinecone, ChromaDB

Operating Systems: Linux, Ubuntu

PROFESSIONAL EXPERIENCE

Machine Learning Engineer

Jan 2025 – Present

STL Digital, Bangalore

- Built and deployed AI-powered applications using LLMs and RAG architectures for enterprise use cases.
- Developed scalable REST APIs using FastAPI for model inference and integration.
- Designed semantic retrieval systems using embeddings and vector databases.
- Fine-tuned open-source LLMs for domain-specific tasks using LoRA and PEFT.
- Deployed machine learning workloads on Google Cloud.
- Implemented CI/CD pipelines for model deployment and automation.
- Improved model performance through inference optimization and prompt engineering.

Key Achievements

- Improved retrieval accuracy by **25%** in RAG pipelines.
- Reduced inference latency by **35%**.
- Reduced deployment cycle time by **40%** through automation.

PROJECTS

Enterprise RAG Chatbot

Python, LangChain, FAISS, FastAPI, GCP

- Built an intelligent document retrieval chatbot using RAG architecture.
- Implemented semantic search using vector embeddings.
- Improved answer relevance and retrieval quality for enterprise documents.

LLM Fine-Tuning Pipeline

PyTorch, Hugging Face, PEFT

- Fine-tuned open-source LLMs on custom datasets for domain adaptation.
- Built automated evaluation workflows for benchmarking model performance.

MLOps Deployment Pipeline

Docker, Kubernetes, GitHub Actions, MLflow

- Designed CI/CD pipelines for training, testing, and deploying ML models.
- Automated model versioning and deployment rollback.

CERTIFICATIONS

- Machine Learning Statistical Foundations Professional Certificate – Wolfram Research
- Programming with Python Professional Certificate – OpenEDG Python Institute
- Python for Data Science Professional Certificate – Anaconda
- Career Essentials in Generative AI – Microsoft
- Career Essentials in GitHub Professional Certificate – GitHub

EDUCATION

Bachelor of Engineering in Information Science & Engineering

2024

Sri Siddhartha Institute of Technology